

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau

Generated on 2026-10-05 18:20

Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	41
Stimuli	1
Observed cells	1714
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96
Cell types	Pluripotency, Extraembryonic_Endoderm, Neuroectoderm, Post-implantation_Epiblast
Network shown	inter_simul.npy, network #0 / 1
UMAP preprocessing	normalise=False, log1p=True

Perturbations (perturbation_simulation)

✓	P1	OV Dnmt3a
✓	P2	OV Col4a2
✓	P3	OV Hoxb2

Contents

1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators + stimulus
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as $NB(k, c / s_i)$ when the depth factor s_i is applied.

Depth

not estimated

Reason

41 genes < 10000

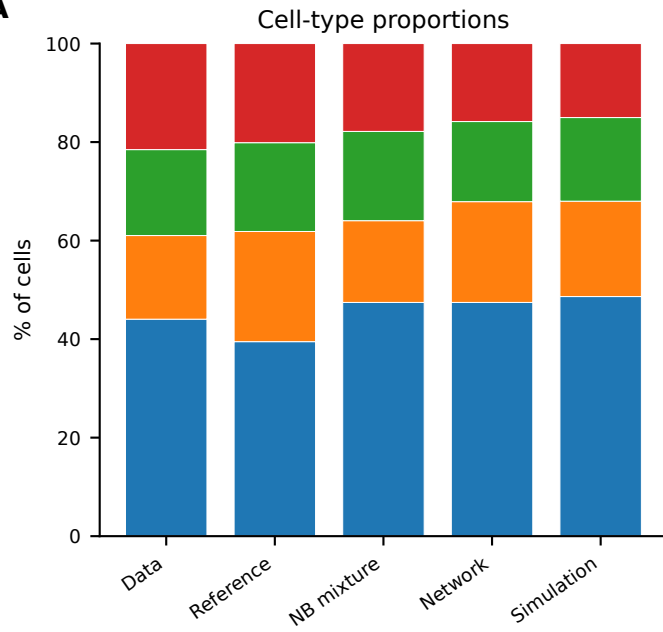
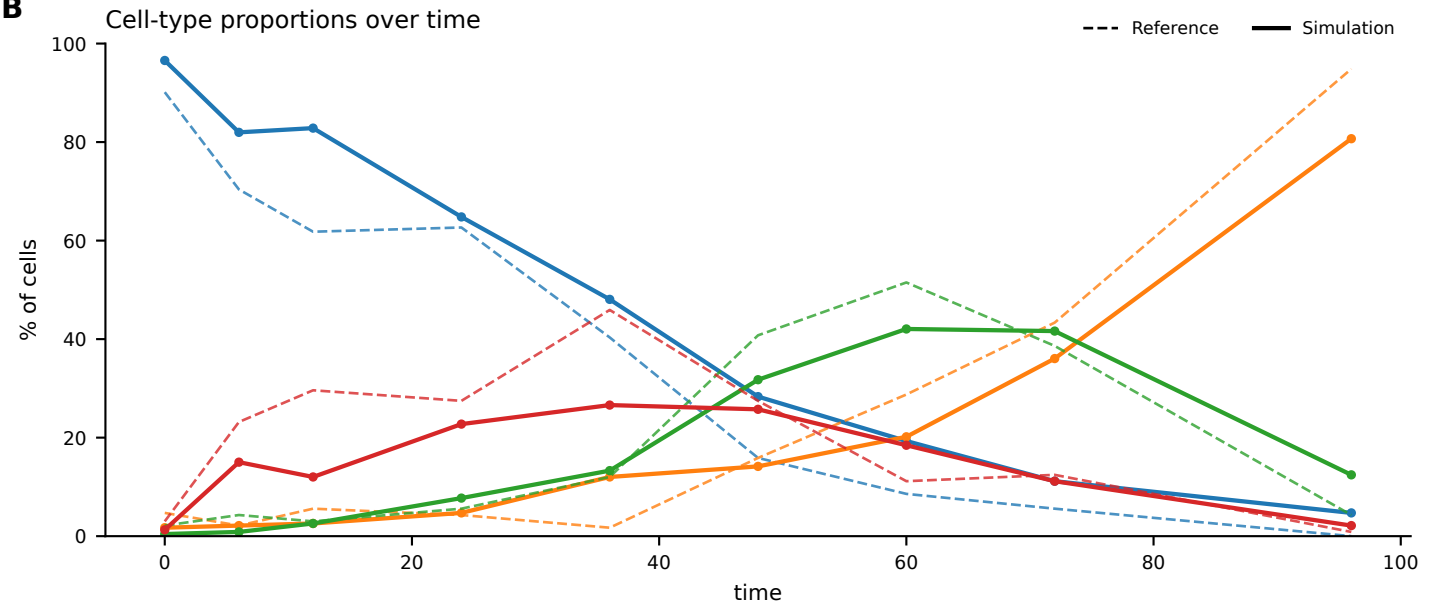
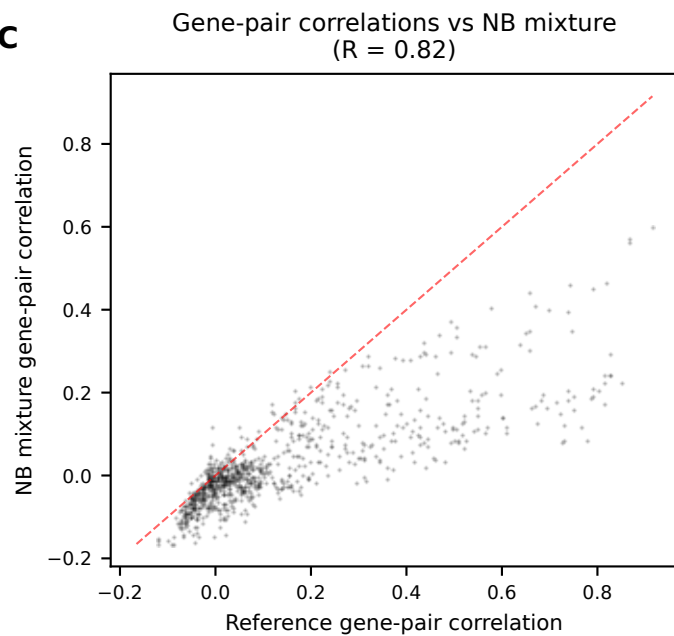
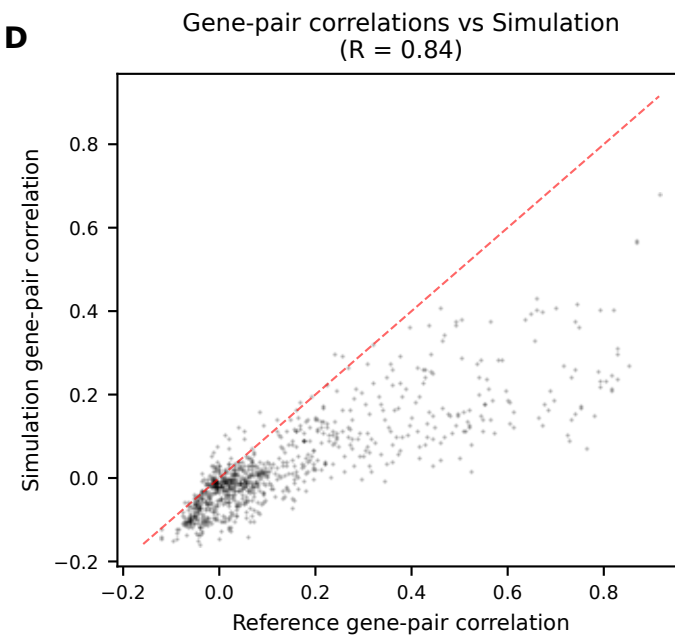
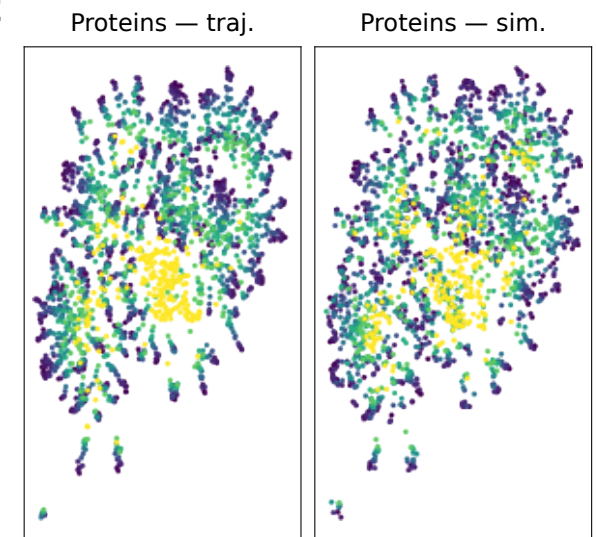
1. Generative model — trajectories and simulation

Joint UMAP of the observed data, the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), NB mixture and network-driven modes along them, and the full simulation.



1. Generative model — quantitative checks

Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

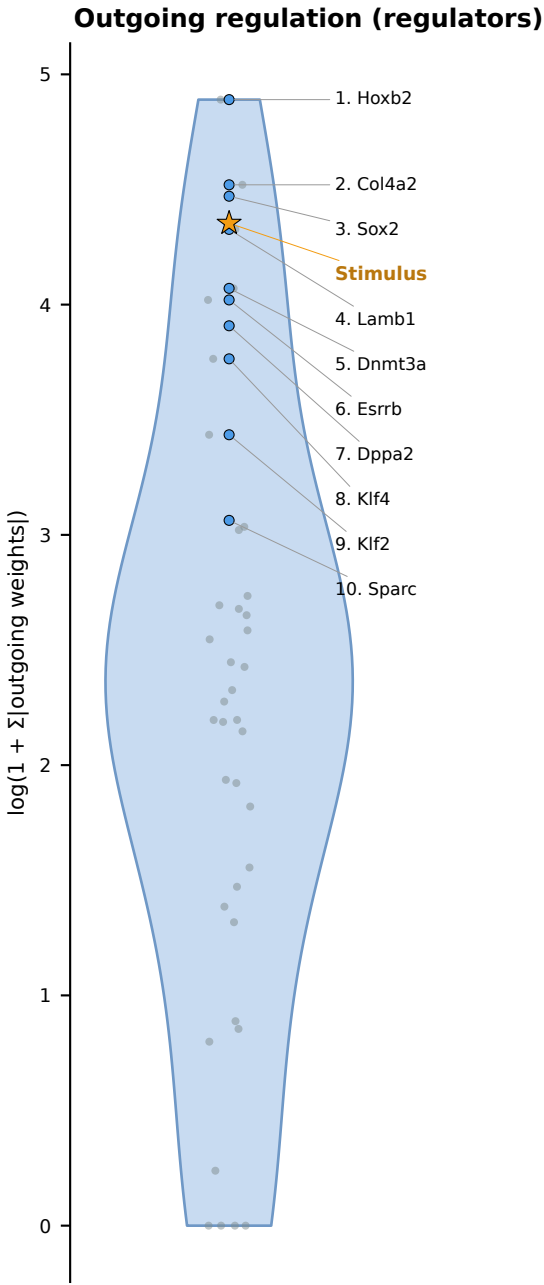
A**B****C****D****E**

Pluripotency Extraembryonic_Endoderm Neuroectoderm Post-implantation_Epiblast

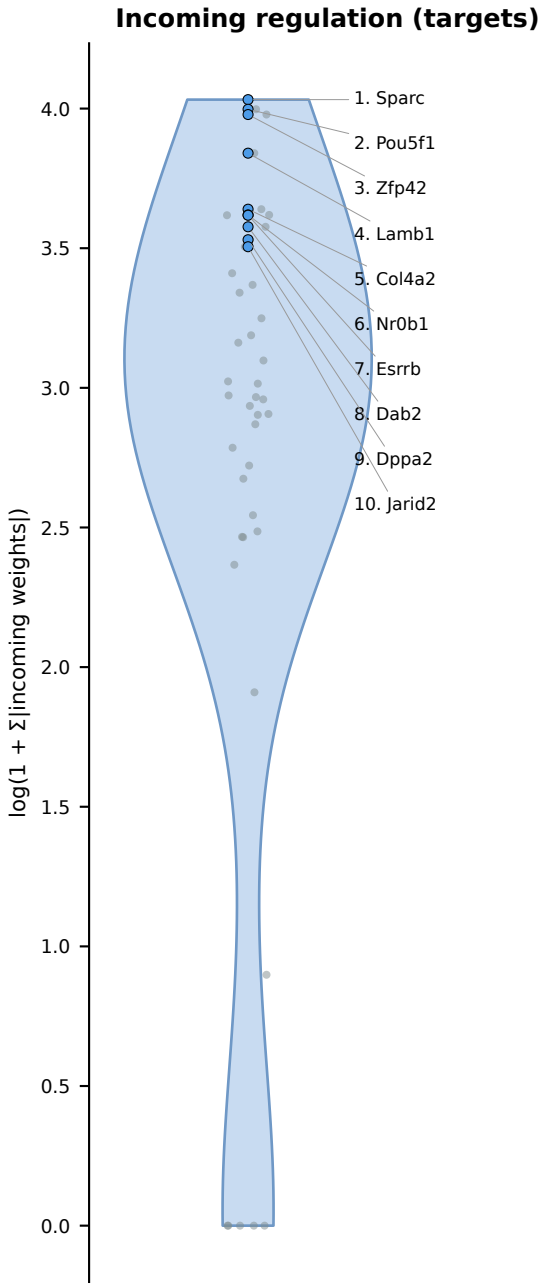
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus shown as a star (stimulus = 1.0 $\geq$ 0.5).

A



B



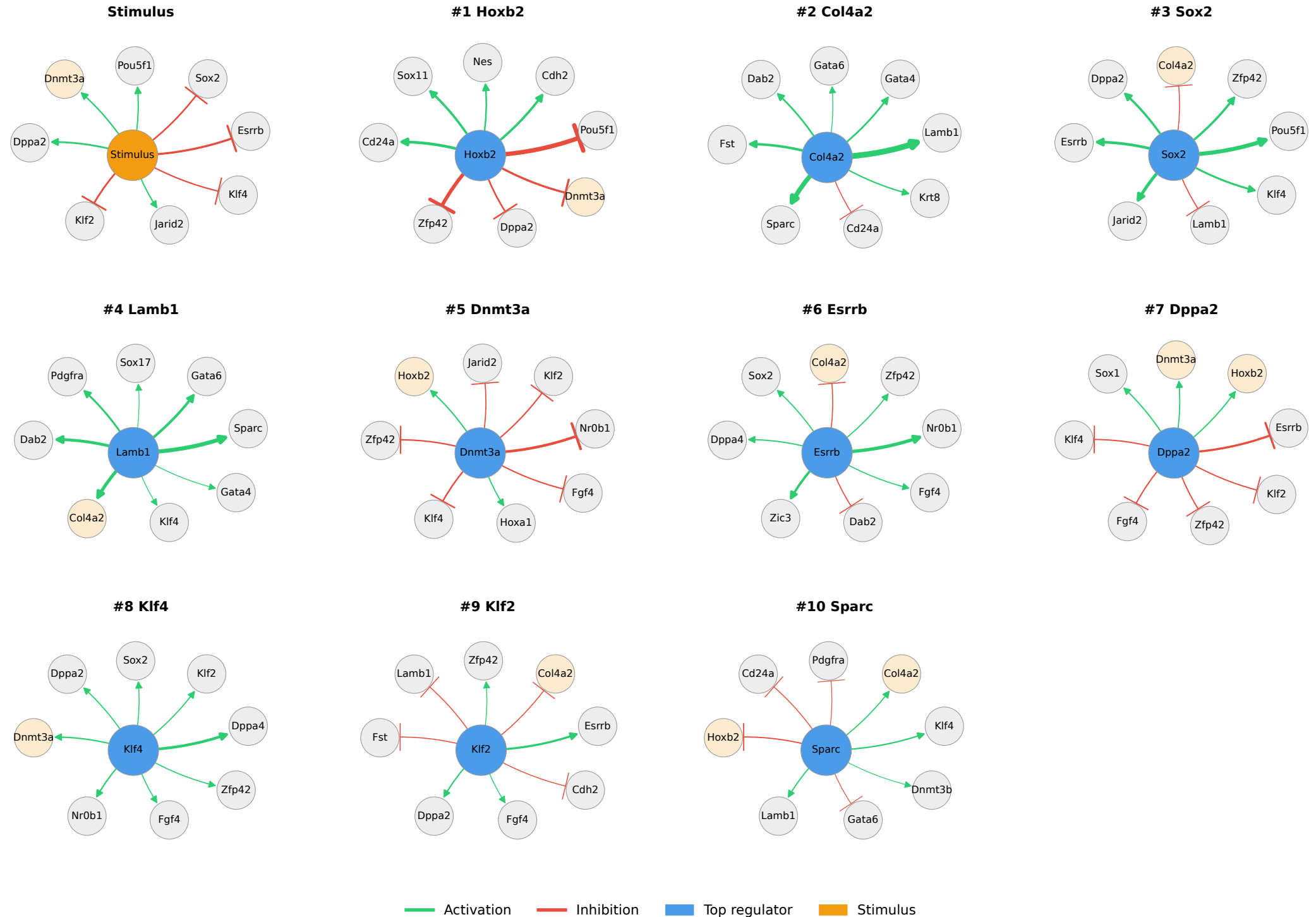
C

Top regulators

#	Regulator	$\Sigma w $	# act.	# inh.
1	Hoxb2	131.99	16	10
2	Col4a2	90.86	12	8
3	Sox2	86.42	13	15
4	Lamb1	74.69	14	6
5	Dnmt3a	57.59	13	20
6	Esrrb	54.72	15	14
7	Dppa2	48.80	16	16
8	Klf4	42.14	16	15
9	Klf2	30.02	15	17
10	Sparc	20.39	15	13
★	Stimulus	76.76	21	9

2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.

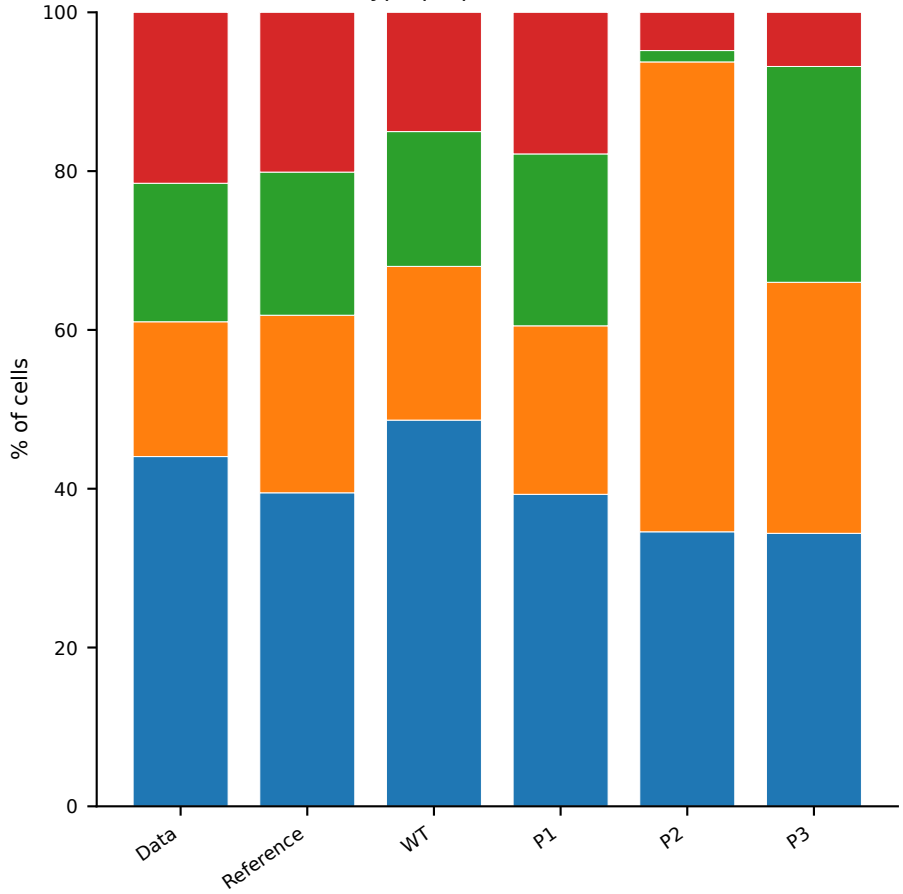


3. In-silico perturbations — overview

Cell types predicted by a random forest trained on the observed data.

A

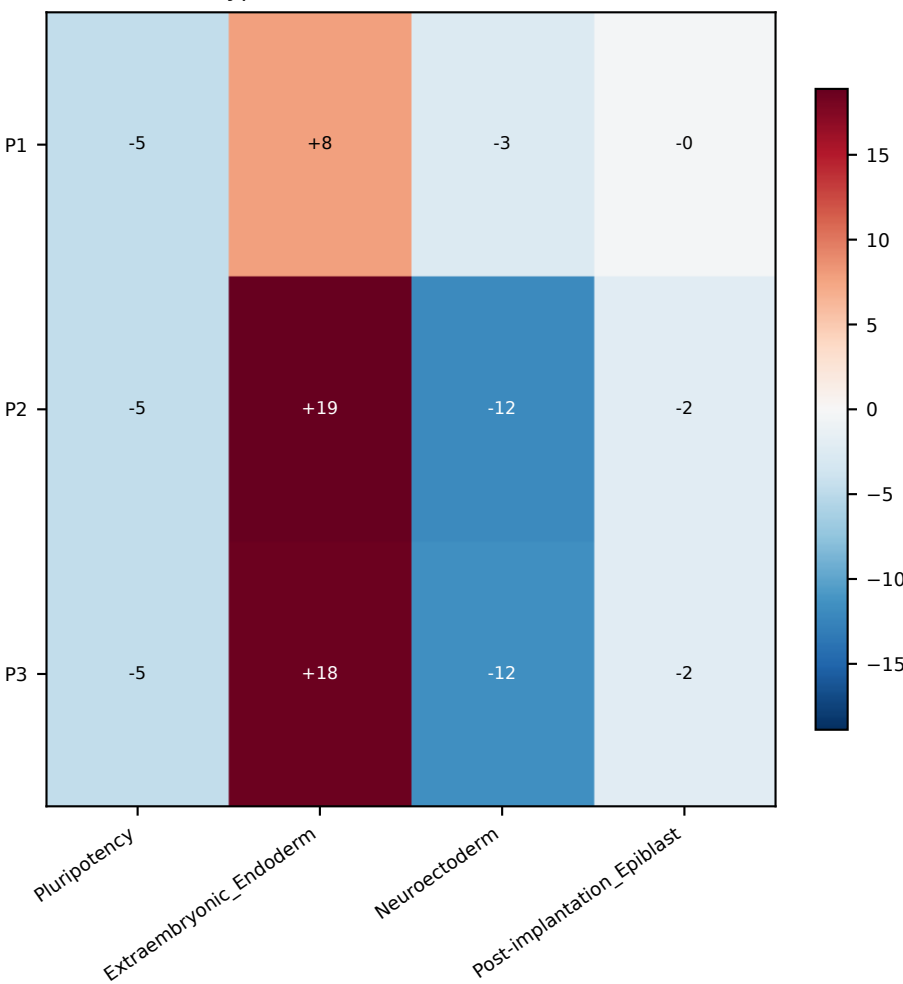
Cell-type proportions (all times)



P1 OV Dnmt3a
P2 OV Col4a2

B

Cell-type shift vs WT simulation at t = 96

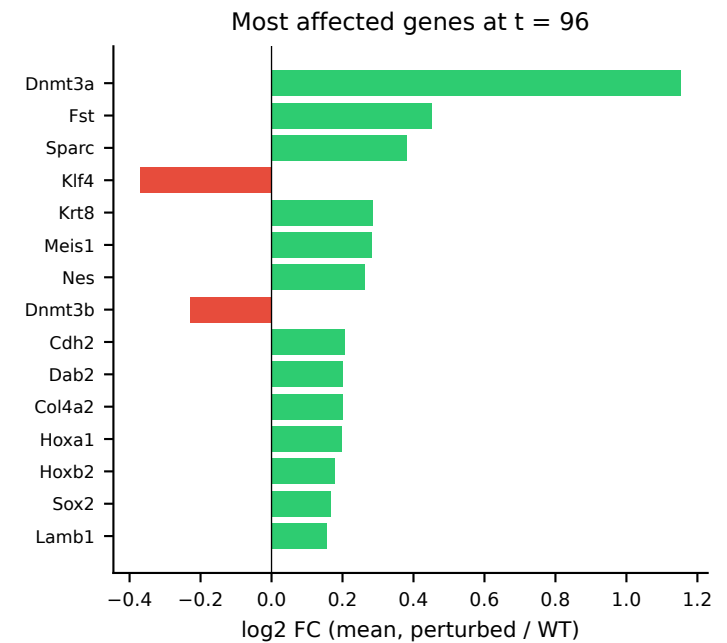
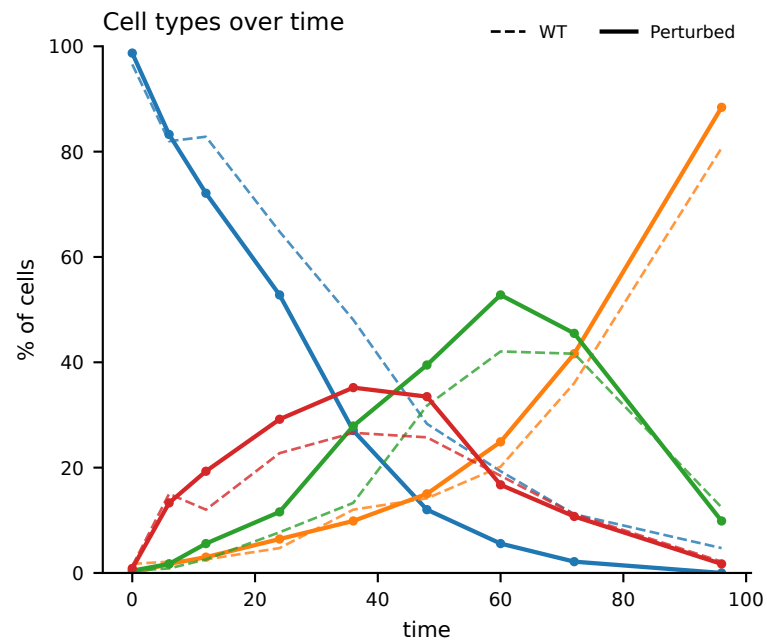
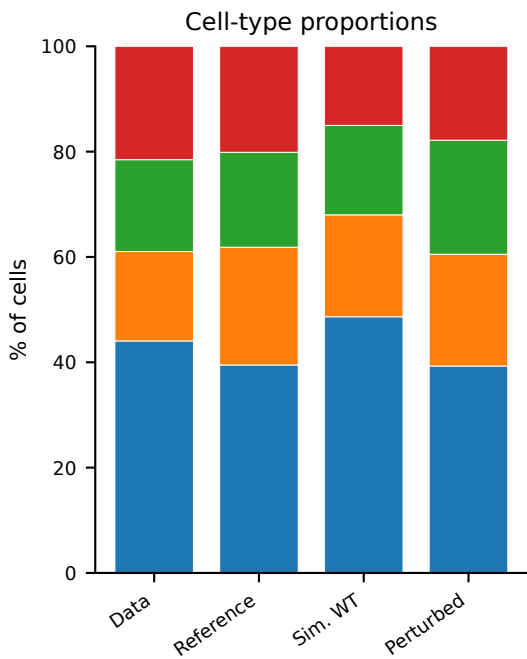
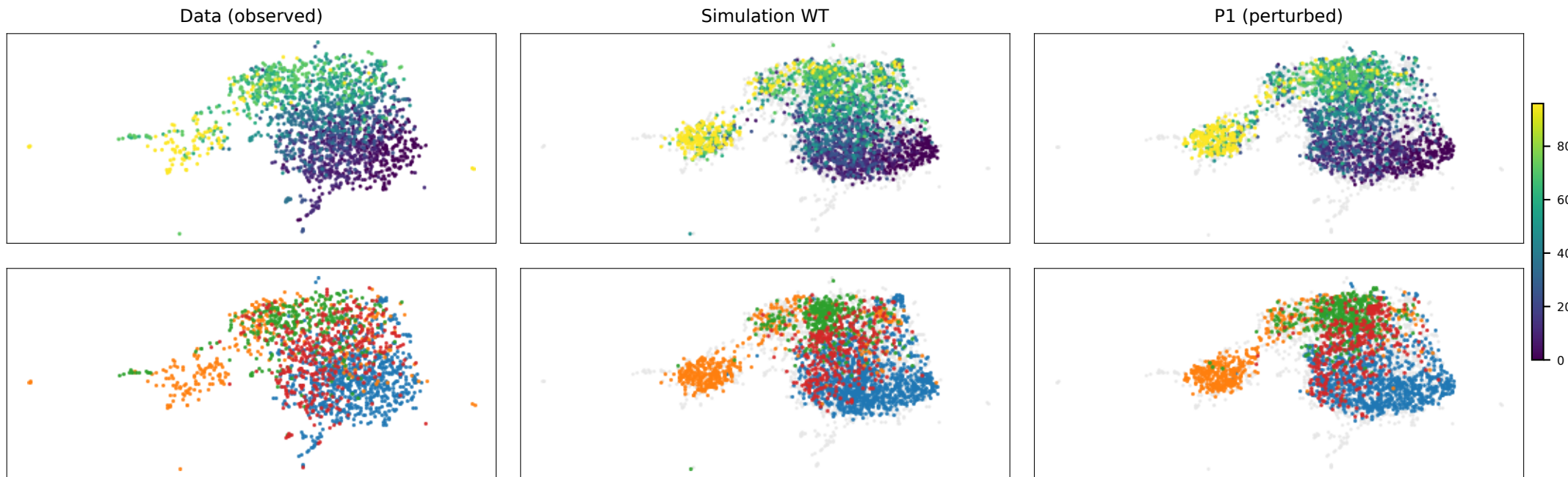


P3 OV Hoxb2

Pluripotency Extraembryonic_Endoderm Neuroectoderm Post-implantation_Epiblast

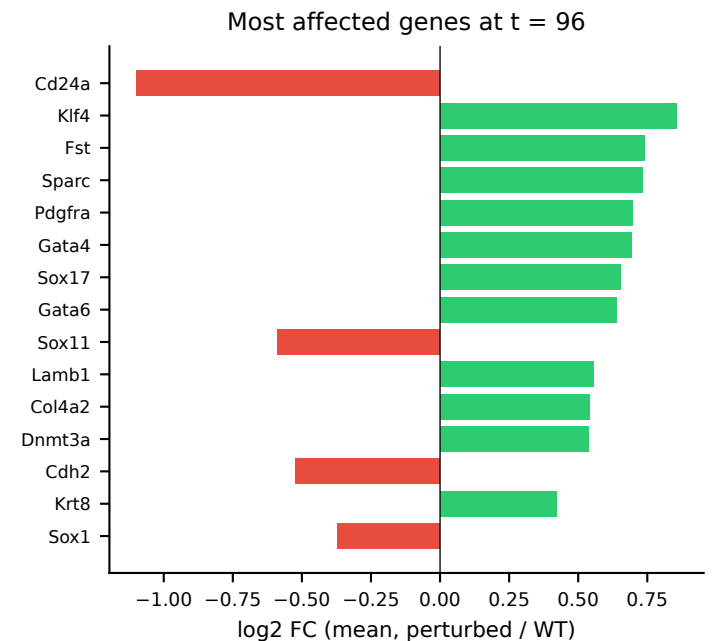
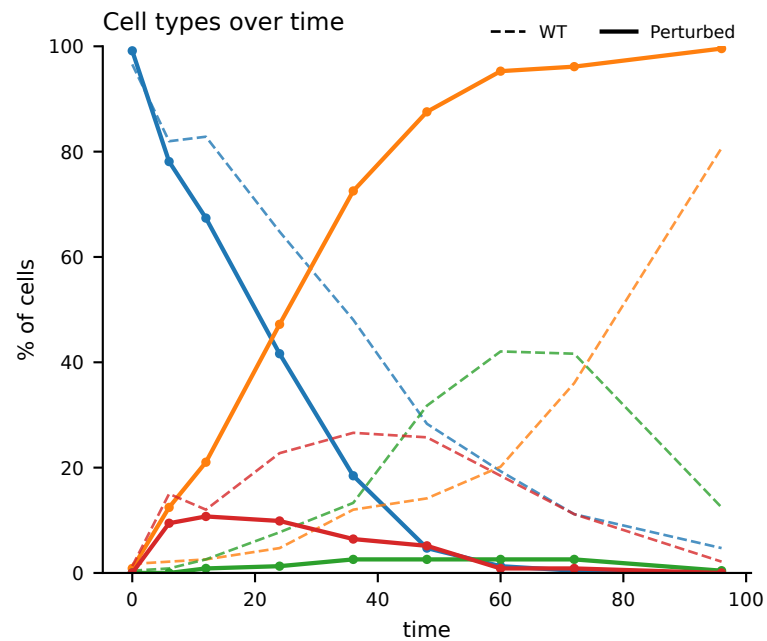
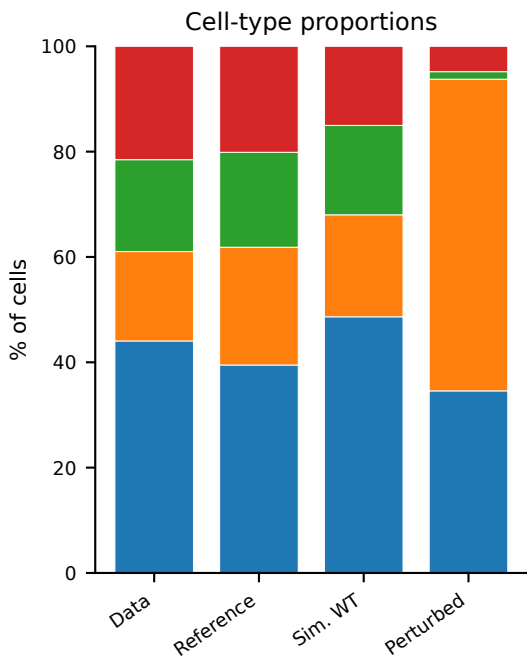
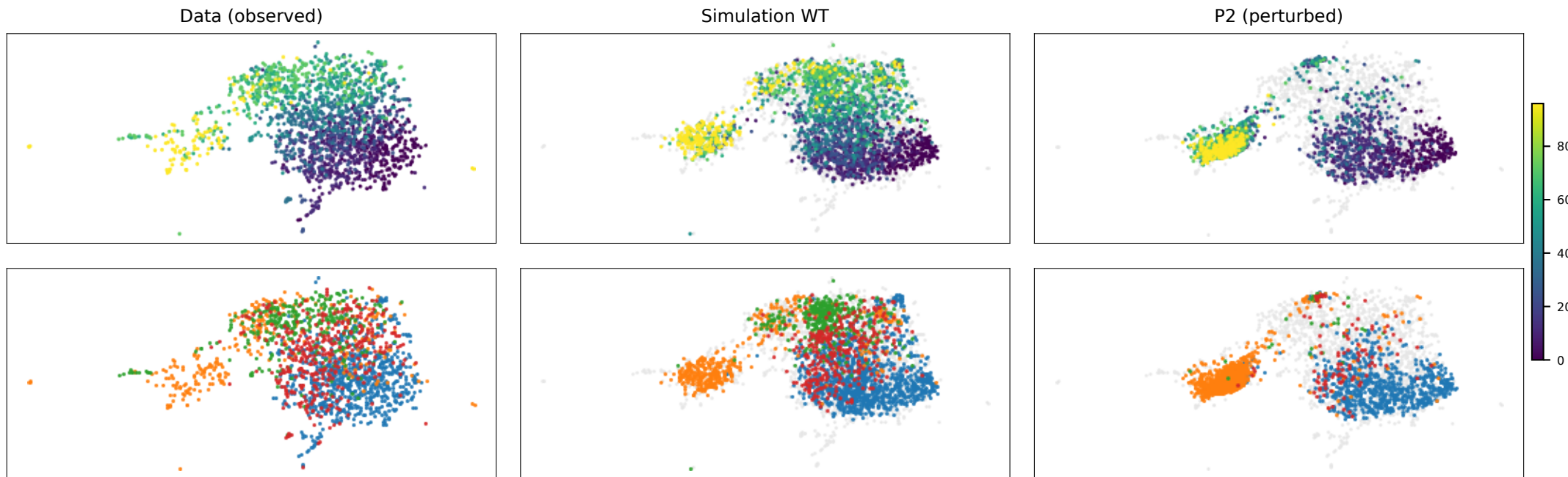
3. Perturbation P1 — OV Dnmt3a

KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



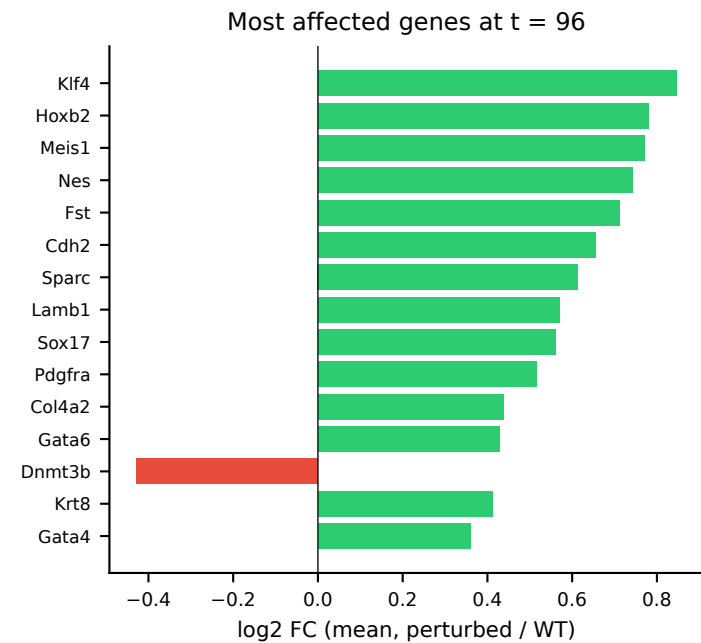
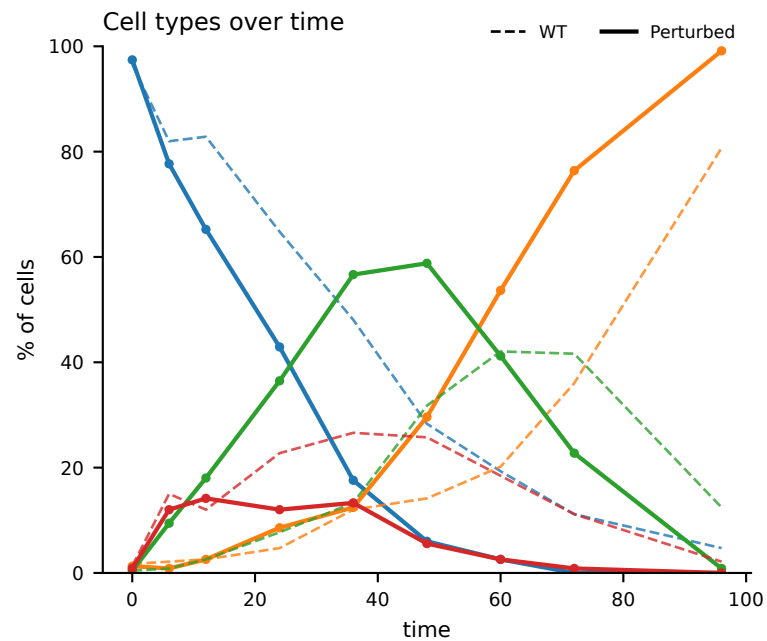
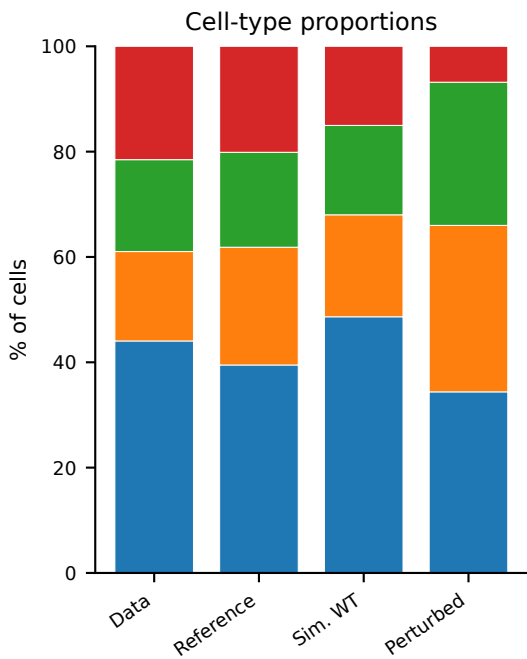
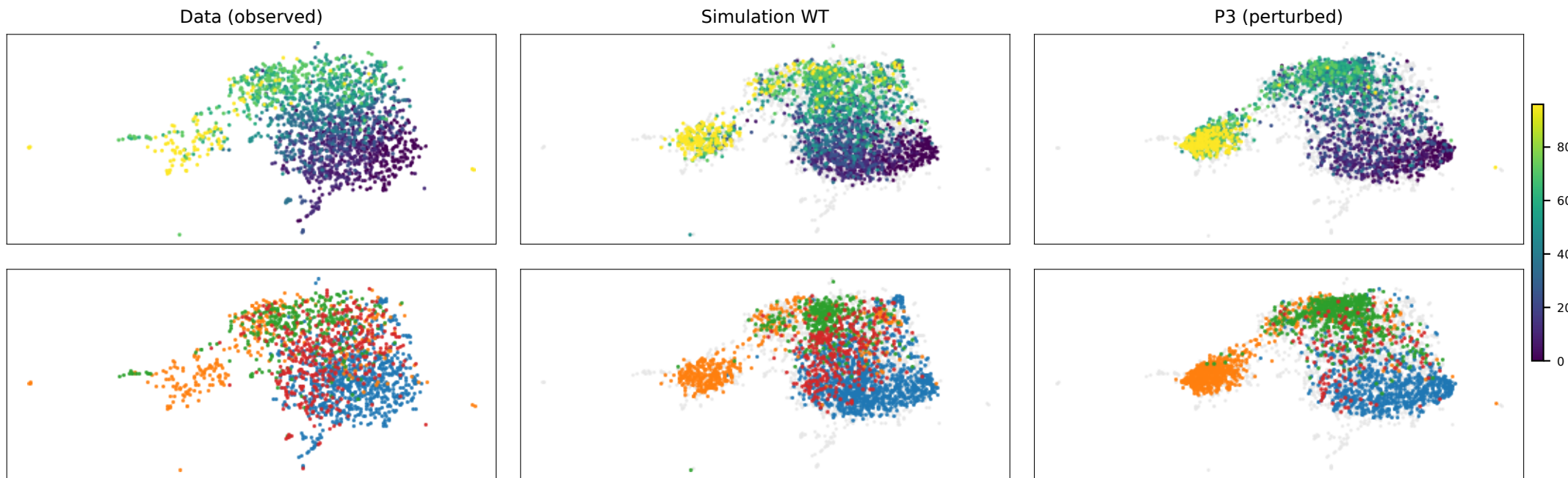
3. Perturbation P2 — OV Col4a2

KO_none_OV_Col4a2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



3. Perturbation P3 — OV Hoxb2

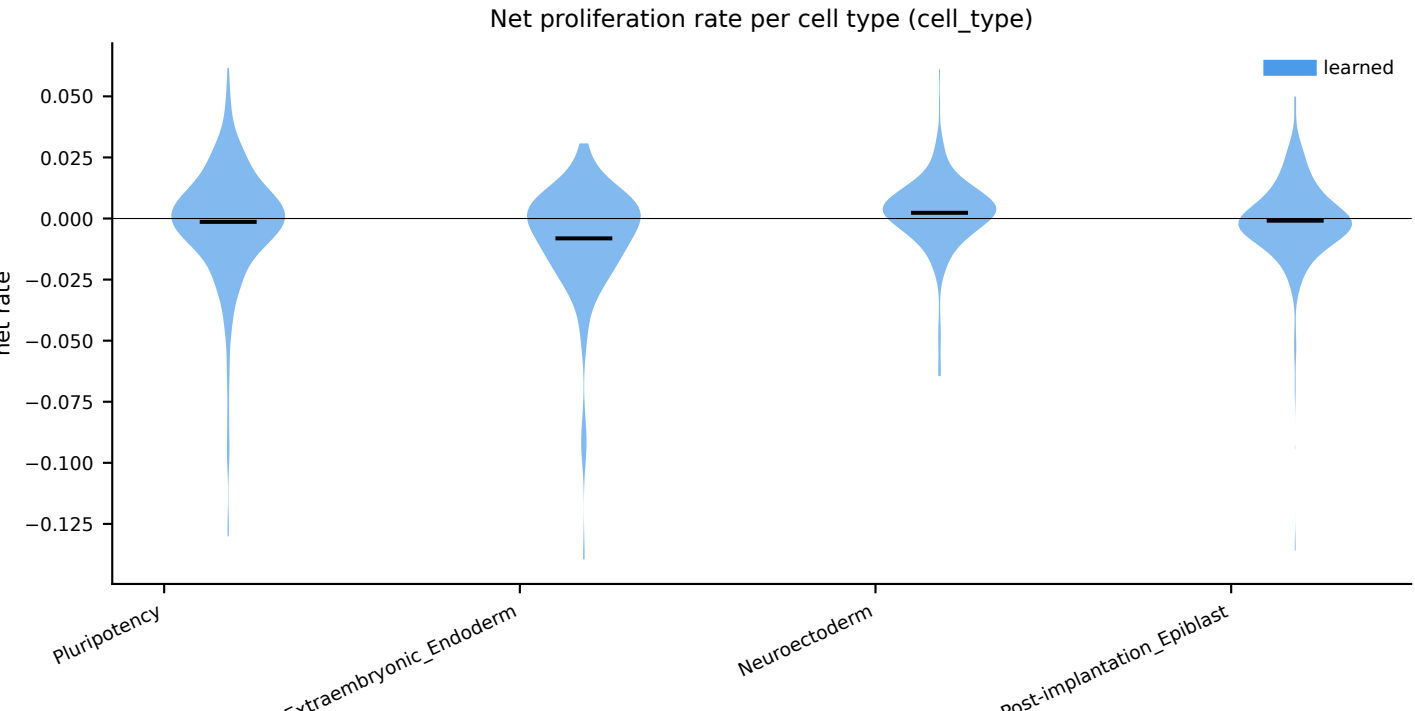
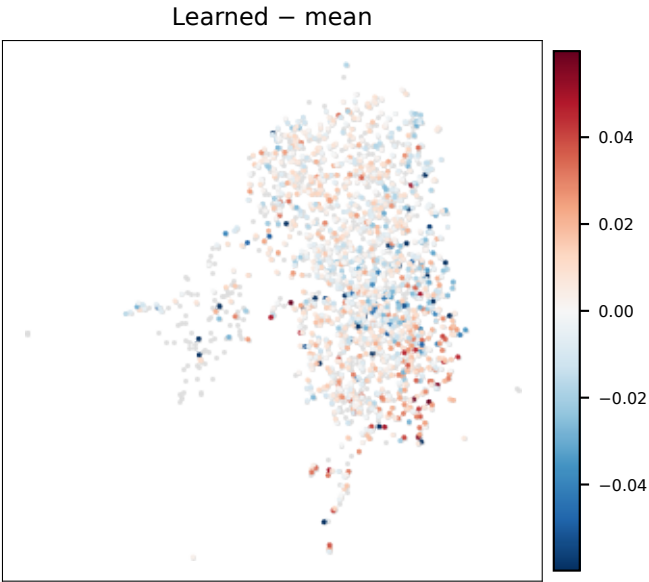
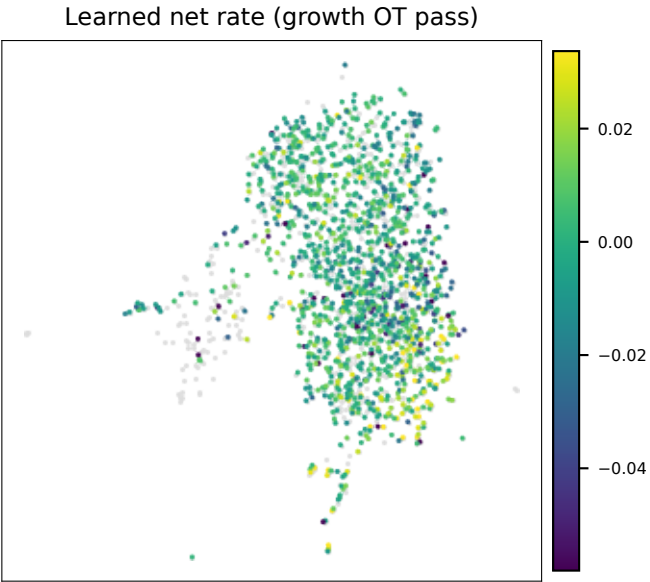
KO_none_OV_Hoxb2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

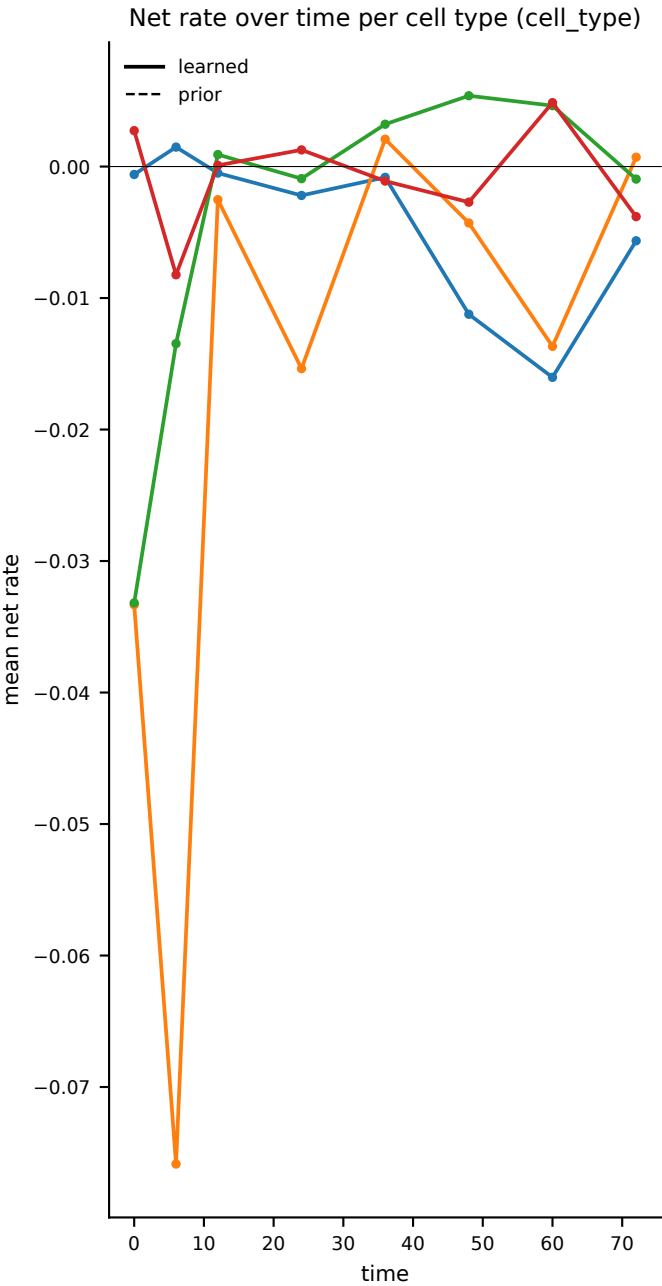
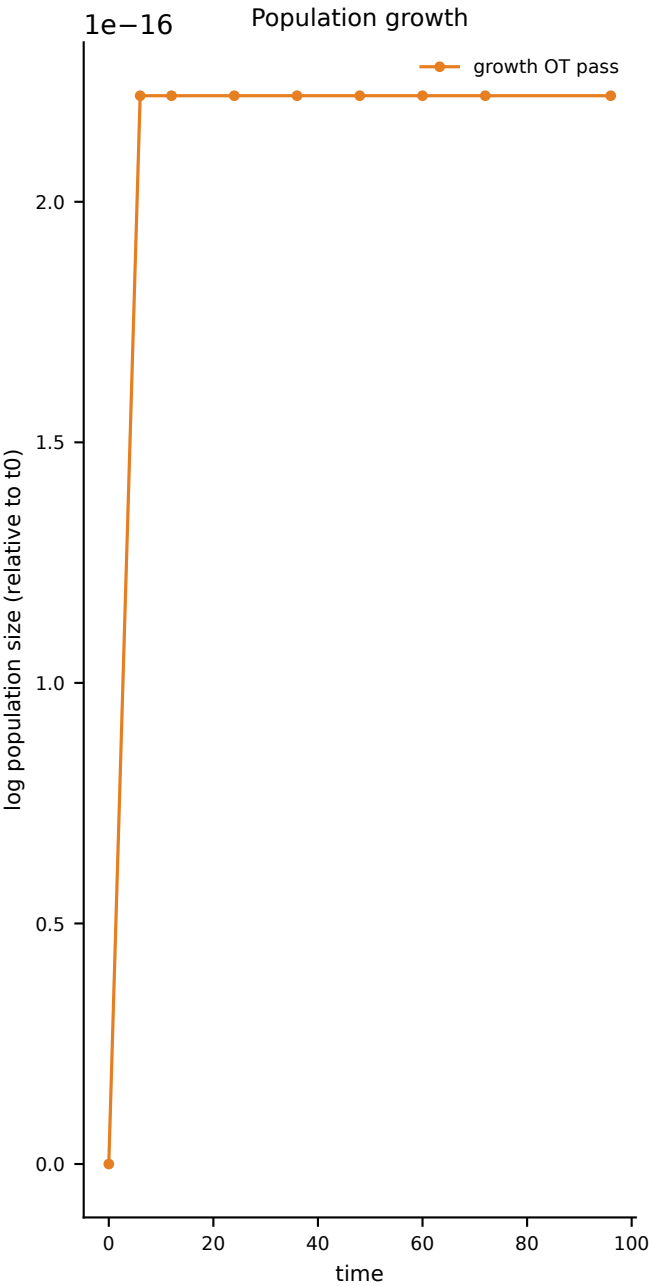


Mean net rate per cell type (cell_type)

Cell type	Anchor	Prior	Learned
Pluripotency	—	—	-0.00137
Extraembryonic_Endoderm	—	—	-0.00813
Neuroectoderm	—	—	0.00234
Post-implantation_Epiblast	—	—	-0.000878

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

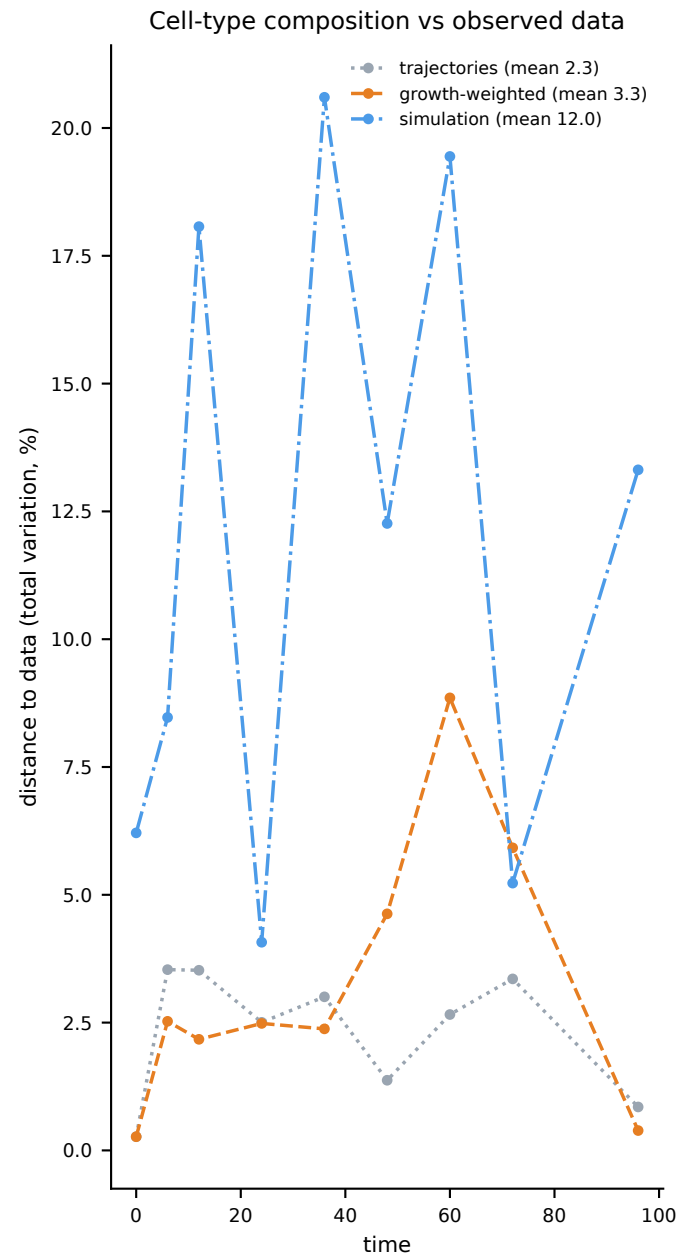
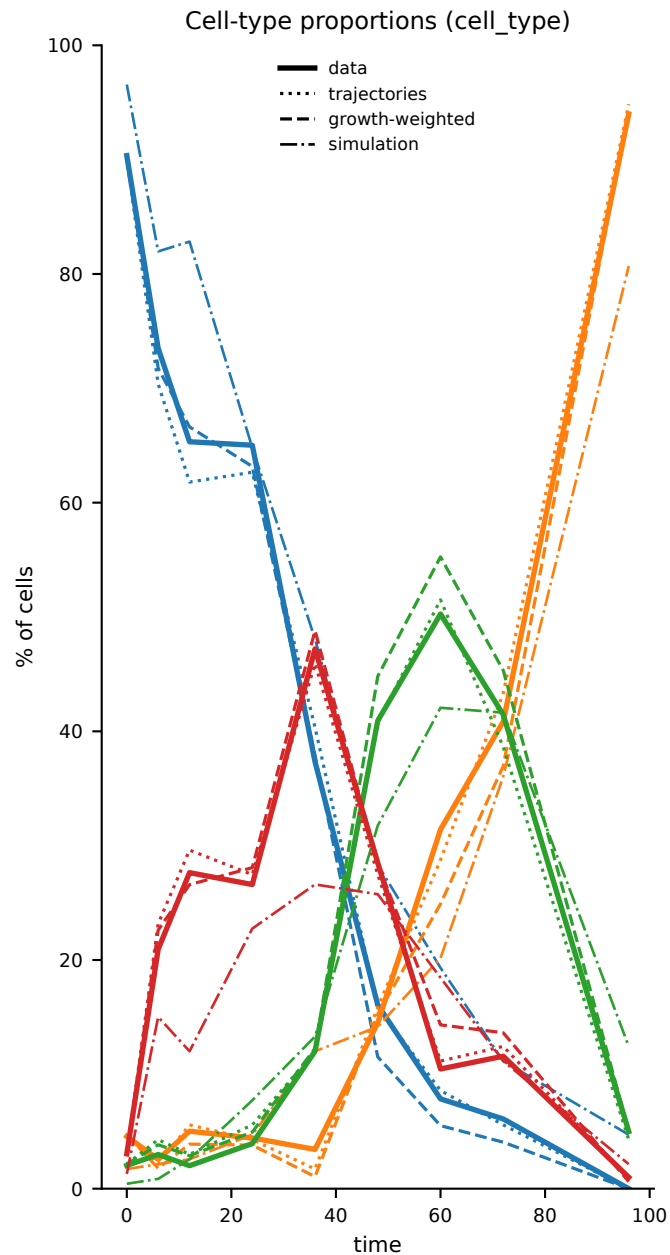


No proliferation MLP
(simulate_with_proliferation = True,
then infer_network_simul)

Pluripotency Extraembryonic_Endoderm Neuroectoderm Post-implantation_Epiblast

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion $\exp(JR_{\text{opt}})$ should bring them back to the observed data.

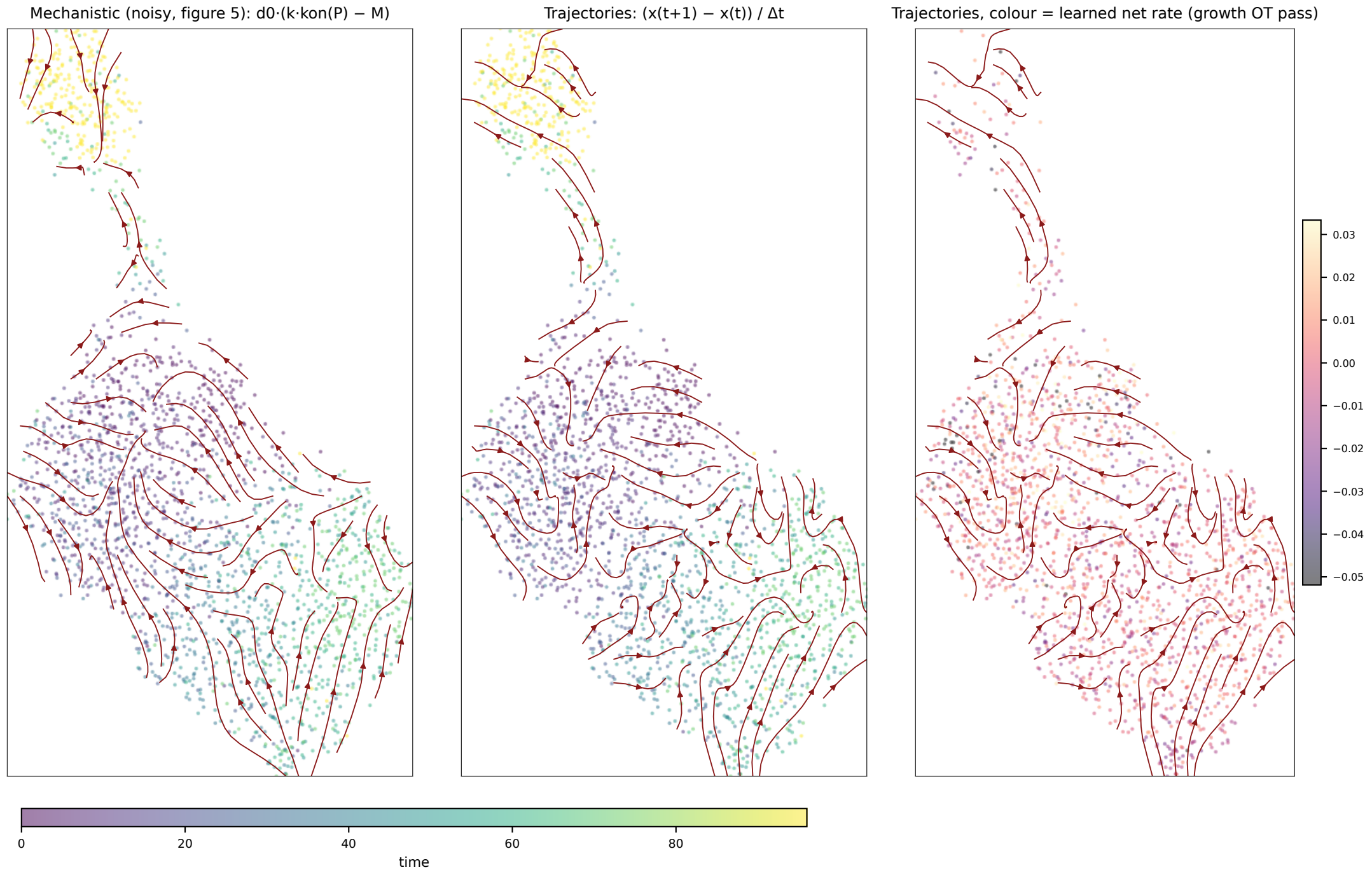


No MLP fit diagnostics
(rerun infer_network_simul
with proliferation)

Pluripotency Extraembryonic_Endoderm Neuroectoderm Post-implantation_Epiblast

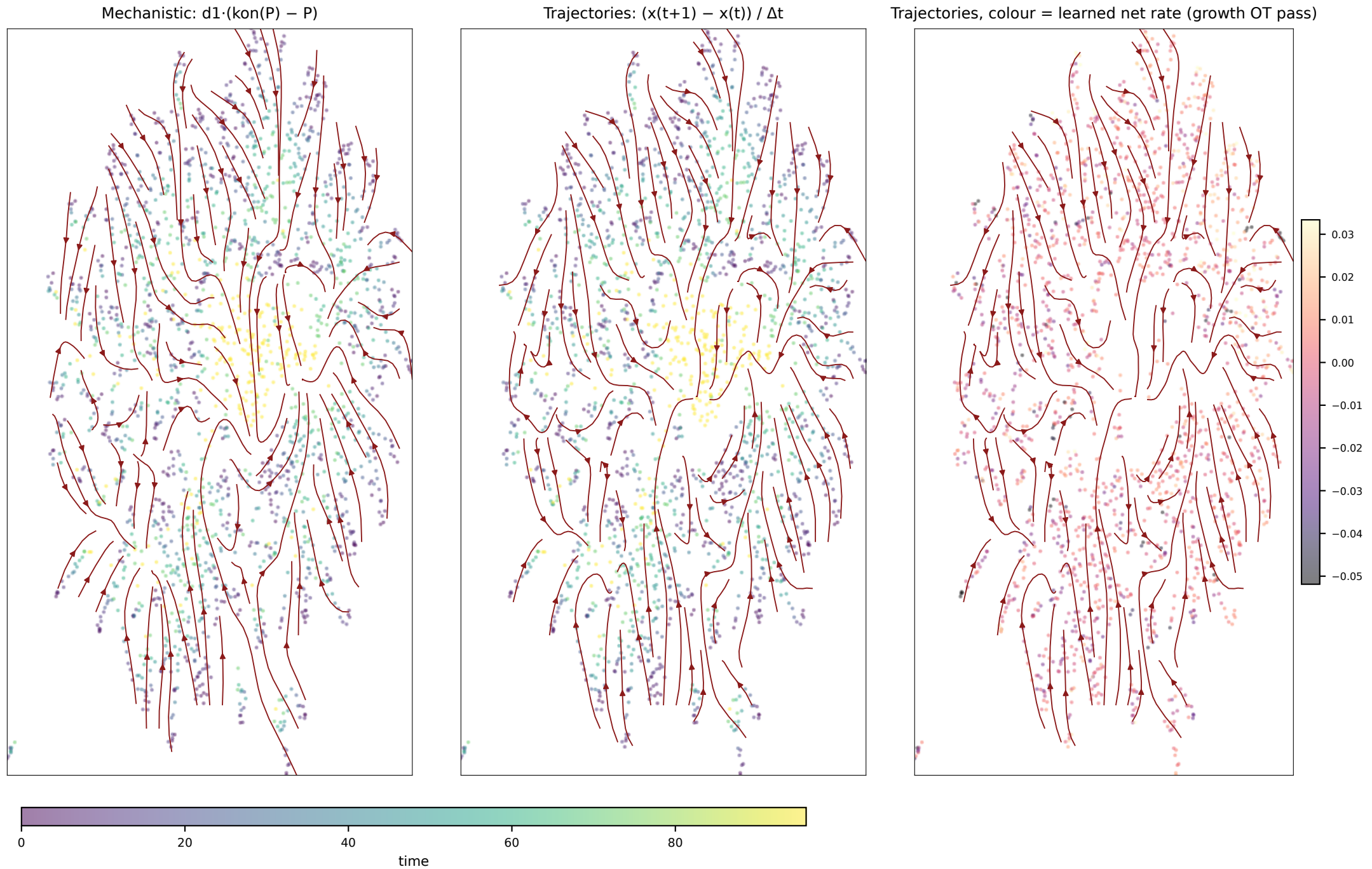
5. Learned dynamics — mRNA velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.59.



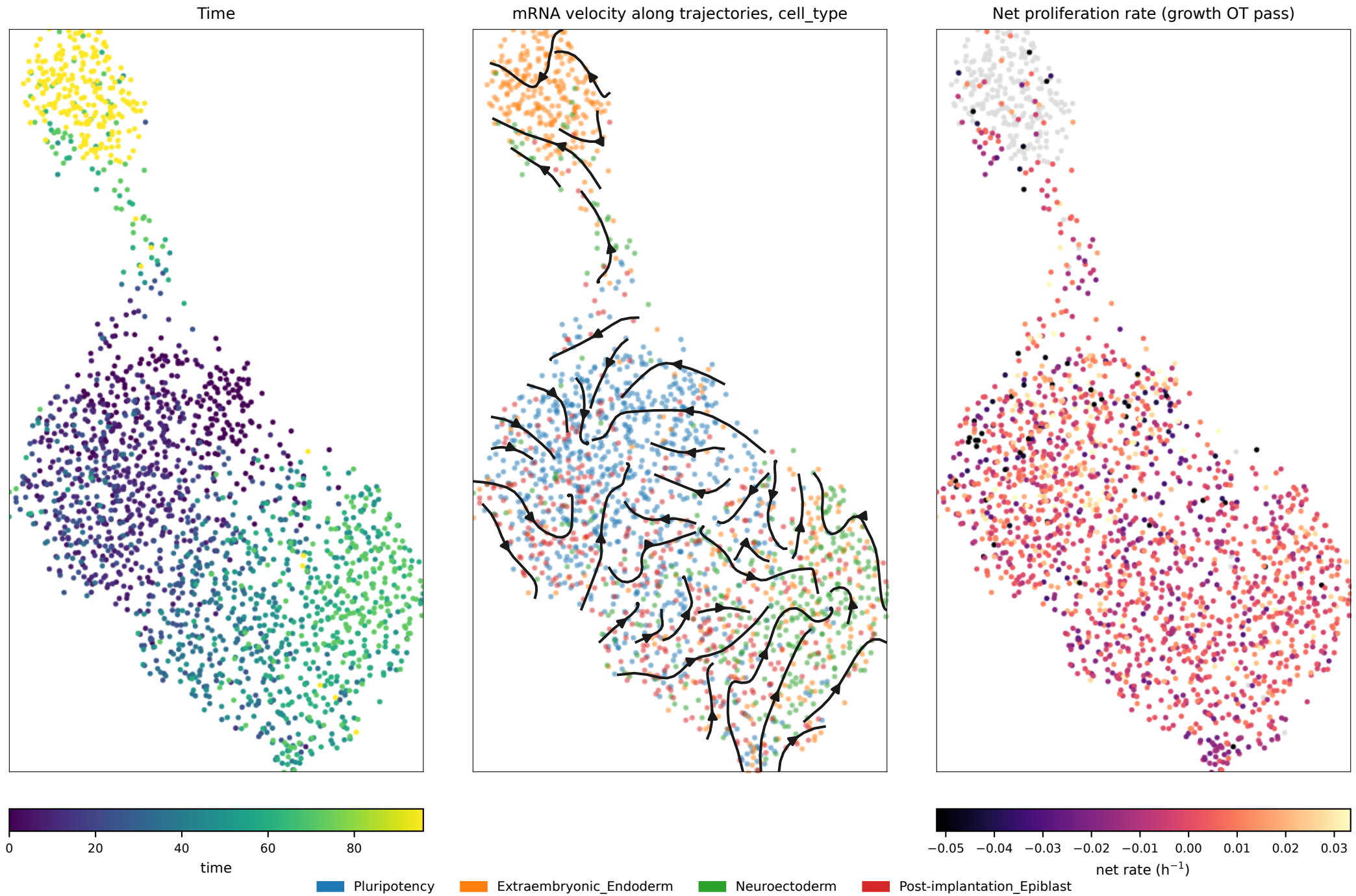
5. Learned dynamics — Proteins velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.71.



5. Learned dynamics — summary

1998 trajectory states on the UMAP of log1p NB-sampled mRNA; velocity along the inferred trajectories $(x(t+1) - x(t)) / \Delta t$; background of the velocity field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the joint UMAP (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

